

Internal Research Report: Do Temporal Network Signals Predict Cross-Disciplinary Concept Spread?

AI Inventor

September 2026

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1 Overview

This document is the internal research report for a five-iteration investigation into whether early structural signals in a scientific concept’s topic co-occurrence network anticipate its later cross-field breadth. The study used the OpenAlex bulk snapshot (476,196,327 works; 2026-09-23) and screened 53 early network indicators across six families against a five-feature popularity baseline on held-out field groups.

The run commissioned 25 artifacts across five iterations; 21 completed and 4 failed or were left incomplete (Table 9). Each iteration below records what was planned, what was built, every result table produced, the reviewer’s verdict, and the hypothesis update that moved the run forward. Dead ends are kept and labelled.

1.1 Panel composition

The analysis panel comprises 12,499 concepts and 27,393 concept-by-field adoption episodes (Table 1).

Table 1: Panel composition. DEV concepts have home fields in Computer Science, Engineering, Biochemistry/Genetics/Medicine. Held-out groups cover the remaining field families. Source: Experiment 5 (iteration 2), `frame_concepts.csv`.

Split	Concepts	Episodes
DEV (CS/Eng/BGM/Med, onset 2003–2009)	4,771	9,079
Cohort (onset 2010–2014, all fields)	4,356	9,799
Held-out Physical Sciences	742	1,662
Held-out Life & Environment	1,113	3,099
Held-out Social Sciences	1,352	3,320
Held-out Math & Decision Sci.	165	434
Total	12,499	27,393

1.2 Outcome: rarefied field breadth

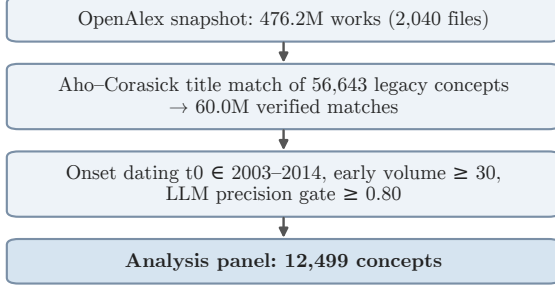
The primary outcome is rarefied field breadth O_{2r} ($m = 50$): the expected number of distinct venue fields among a fixed-size random draw of m papers from a concept’s publications in years $t_0 + 6$ to $t_0 + 8$, computed by exact hypergeometric rarefaction.

2 Iteration 1: Wide screen of rival spread signals

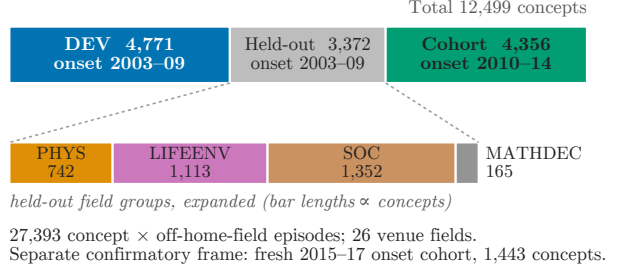
2.1 Strategy and rationale

Iteration 1 ran a wide screen across five rival mechanisms for why a new concept becomes broadly integrated. The five mechanisms were: (1) lineage naturalisation (background-adjusted A_h^*), (2) social reach through unconnected author groups, (3) structural diversity of new co-occurrence neighbours, (4) frequency-free selectivity, and (5) landing in gateway fields plus adopter insularity. All were screened on one frozen dev panel (48 concepts, CS/Eng/BGM/Med homes, onset 2003–2009), with one baseline, one outcome set and one pre-registered rule.

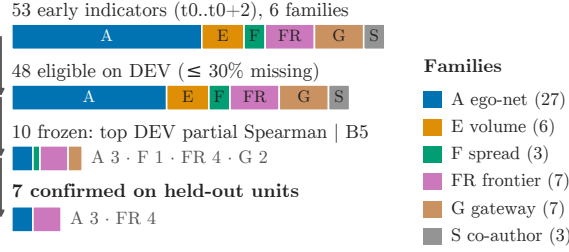
(a) Concept identification



(b) Panel structure



(c) Indicator screen (breadth outcome)



(d) Research questions

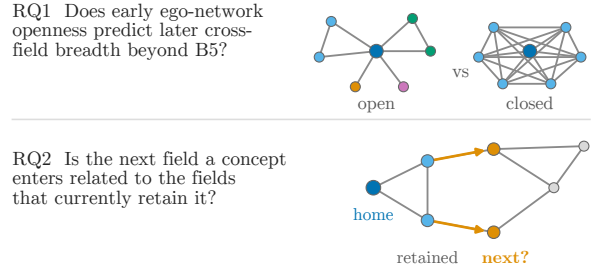


Figure 1: Overview of the study design. (a) Concept identification: 56,643 legacy OpenAlex concepts are matched by title (Aho–Corasick) against a 476.2M-work OpenAlex snapshot, giving 60.0M verified matches; concepts are onset-dated ($t_0 \in 2003-2014$), required to have early volume ≥ 30 and to pass an LLM precision gate ≥ 0.80 , yielding an analysis panel of 12,499 concepts. (b) Panel structure, bar lengths proportional to concept counts: DEV (blue; CS, Eng, BGM and Med homes, onset 2003–09) 4,771, held-out field groups (grey; onset 2003–09) 3,372, and the 2010–14 onset cohort (green) 4,356. The expanded lower bar splits the held-out block into PHYS 742, LIFEENV 1,113, SOC 1,352 and MATHDEC 165. The panel contains 27,393 concept $\times$ off-home-field episodes over 26 venue fields; a separate fresh 2015–17 onset cohort of 1,443 concepts serves as a confirmatory frame. (c) Indicator screen for the breadth outcome (O2r_m50). Bars are coloured by indicator family (A ego-net, E volume, F spread, FR frontier, G gateway, S co-author) with lengths proportional to indicator counts: 53 early indicators computed over $t_0..t_0+2$; 48 eligible on DEV (the 5 dropped D-family ego-net indicators were more than 30% missing); the 10 frozen on DEV by partial Spearman correlation given the five-feature baseline B5 (A 3, F 1, FR 4, G 2); and the 7 confirmed on held-out units, the four held-out groups plus the two parts of the 2010–14 cohort (A 3, FR 4). (d) The two research questions, drawn as schematic illustrations with no data. RQ1 asks whether an open early ego network, as opposed to a closed one, predicts later cross-field breadth beyond B5. RQ2 asks whether the next field a concept enters (orange) is related to the off-home fields that currently retain it (light blue), starting from its home field (dark blue) on the field backbone.

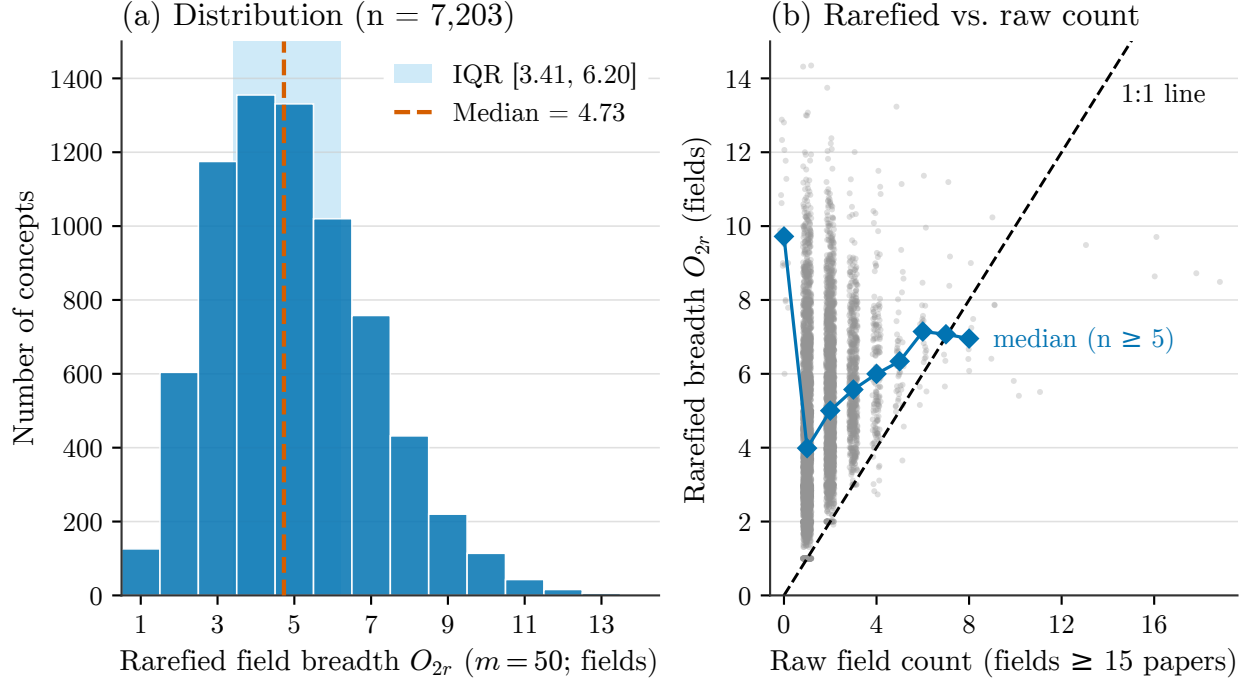


Figure 2: Distribution of the primary outcome, rarefied field breadth (O_{2r} , $m = 50$), over the 7,203 of 12,499 frame concepts that have at least 50 papers in the $t_0+6..t_0+8$ outcome window (O_{2r} is undefined below that). (a) Histogram of O_{2r} in 1-field bins centred on the integers. The dashed orange line marks the median (4.73) and the light-blue band the interquartile range [3.41, 6.20]. The distribution is mildly right-skewed: most concepts reach 2–8 fields and a thin tail extends to 14. (b) O_{2r} against the raw field count, the number of venue fields holding at least 15 of the concept’s outcome-window papers. Grey dots are concepts (x jittered), blue diamonds are per-count medians (counts with $n \geq 5$) and the dashed black line is the 1:1 line. Rarefied breadth lies above the 1:1 line for 97.6% of concepts and rises only weakly with the raw count (Spearman $\rho = 0.31$). It is nearly independent of outcome-window volume ($\rho = -0.06$), whereas the thresholded raw count grows with volume ($\rho = 0.41$). Rarefaction thus separates size-adjusted breadth from volume-driven field counts.

The shared OpenAlex credit pool ran dry during this iteration (139 own credits spent by Experiment 1), forcing data deviations: yearly counts used OpenAlex S0 exactly, but field labels and citation lineage came from free Semantic Scholar data.

Five artifacts were commissioned; three completed, and two failed.

2.2 Experiment 1: Does citing a concept ‘as your own’ predict spread?

Screen of candidate L , the background-adjusted naturalisation gap A_h^* , on 48 dev concepts. The pre-registered rule was not passed:

Table 2: Experiment 1 screen result for A_h^* (iteration 1). Source: `screen_result.json`.

Clause	Value	Pass?
$\Delta\rho$ for O_{2r} over B5	-0.006 $[-0.034, +0.017]$	No
Positive left-out groups $\geq 3/4$	0 of 4	No
Split-half reliability (Spearman–Brown)	0.58	No
$ \rho $ with log early vol / early growth ≤ 0.6	0.14 / 0.18	Yes

Additional findings from Experiment 1:

- A_h^* ’s within-field Spearman with O_{2r} flipped sign: Medicine $+0.45$, CS -0.18 .
- Field-level $\rho_{cj}^* \rightarrow R_j$: $\Delta\text{AUC} +0.002$ $[-0.011, +0.016]$ over 367 episodes.
- **M1 background homophily**: R^2 of raw lineage log-OR on background log-OR = 0.66; background log-OR was positive for 48/48 concepts. Two-thirds of between-concept variance in raw lineage autonomy is general disciplinary homophily, not concept-specific integration.
- REML variance components: $\tau_c = 0.29$, $\tau_{cj} = 0.65$, indicating that most variation in cross-field behaviour sits at the concept $\times$ field level, not the concept level.

2.3 Experiment 3: Do diverse topic ties predict concept spread?

Screen of two co-occurrence emergence indicators on 47 dev concepts, using a full-corpus topic PMI backbone per slice (Leiden $\gamma = 3$, giving 25/26/23 communities). D_z failed the size diagnostic (ρ with log volume = -0.63), so the pre-declared fallback D_{ratio} was primary.

Table 3: Experiment 3 screen results (iteration 1). Source: `screen_result.json`.

Indicator	$\Delta\rho$	90% CI	Groups pos.	SB
D_{ratio}	$+0.006$	$[-0.092, +0.135]$	3/4	0.83
F_{res}	-0.060	$[-0.158, +0.014]$	1/4	0.44

Neither survived the pre-registered rule. However, an exploratory out-of-group partial ρ of D_{ratio} given B5 was $+0.335$ $[+0.02, +0.65]$, permutation $p = 0.037$, suggesting the signal was absorbed by the high B5 baseline ($\rho_{B5} = 0.770$) rather than absent.

The experiment also computed about 30 indicators across the ego-network. A portable negative signal, edge persistence, showed within-group ρ of -0.34 , -0.52 , -0.07 , -0.22 across all four groups.

2.4 Experiment 4: Where a concept lands early vs how broadly it spreads

Screen of candidate G (gateway landing) on 46 dev concepts (34 with an outcome window). This artifact also produced the authoritative shared outcome tables.

Table 4: Experiment 4 screen results (iteration 1). Source: `screen_result.json`.

Metric	Value	CI
$\Delta\rho$ for O_{2r} (B5+G vs B5)	+0.033	$[-0.095, +0.168]$
Positive groups	2/4	–
Reliability (r_{SB})	0.92	–
$O_{2r_resid} \Delta\rho$	+0.15	$[+0.000, +0.321]$
O1 ΔAUC	+0.072	$[+0.00, +0.16]$
Field-level gateway retention ΔAUC	+0.103	$[+0.03, +0.17]$

G did not survive the pre-registered rule for the primary O_{2r} outcome but showed a field-level lead for retention: adding the adopting field’s eigenvector gateway centrality raised retention AUC from 0.705 to 0.808 (+0.103). Next-field entry: relatedness density AUC 0.61, beaten by log field size (0.74).

2.5 Failed artifacts (Dead ends)

Two artifacts failed due to REPL timeout:

- **Dataset 1** (sealed held-out Frame-N test set): timed out. Consequence: no held-out evaluation existed after iteration 1.
- **Experiment 2** (candidate S, unconnected co-author groups, Cheng et al. 2023): timed out. Candidate S was untested (not refuted) until iteration 3.

2.6 Review and hypothesis update

The reviewer scored iteration 1 at 3 (blocking). Key critiques: several conclusions contradicted artifact evidence (B5 baseline ρ was only 0.327 for Experiment 4, not 0.77–0.83 as claimed; the A_h^* medians were actually within-group correlations; “strongest secondary signal” was false). Two failed artifacts were omitted entirely.

The hypothesis update moved to **deepen**: the best strand was a lead (gateway retention ΔAUC +0.10, dev only, $n = 80$). The unit of analysis shifted to the concept×field episode (justified by $\tau_{cj} = 0.65 > \tau_c = 0.29$). A_h^* and D_{ratio} were closed as headline bets. Confidence decreased.

3 Iteration 2: Testing gateway retention at scale

3.1 Strategy and rationale

Iteration 2 moved the budget from more metrics to more samples, as the reviewer demanded. One zero-credit scan of all 2,040 OpenAlex S3 works files produced a 12,499-concept panel with 27,393 episodes and a hash-sealed held-out split. Five artifacts were commissioned, all completed.

3.2 Experiment 5: Sealed held-out test of gateway retention

Built the authoritative S1 frame tables. The hash-sealed spec was unsealed once.

H1 (gateway retention) — DISCONFIRMED. Held-out $\Delta\text{AUC} = -0.00001$ $[-0.0006, +0.0003]$; DL pooled -0.00004 ($I^2 = 0$). The baseline ladder showed gateway’s DEV signal ($+0.0019$) vanished once the leave-concept-out field propensity $P_j(-c)$ was added, and reversed on held-out data. The shuffled- R placebo’s 95th percentile (0.130) exceeded the original $+0.103$, so the iteration-1 lead could not be certified as above chance on 80 episodes. Minimum detectable ΔAUC was 0.004.

H3 (concept-level gateway landing): Held-out partial ρ : $G = 0.030$, $G_A = 0.026$, $G_{\text{btw}} = 0.046$ (Holm $p = 0.0045$); DL pooled $G = 0.068$ $[0.029, 0.107]$. However, the concept-bootstrap CI of pooled G included zero $([-0.006, +0.065])$, and DEV-to-held-out shrinkage was from 0.14 to 0.03.

3.3 Experiment 6: Where new scientific concepts spread next

Conditional logit on 653 newborn concepts (1,865 episodes). Relatedness to retained fields predicted the next field entered: held-out $d_0 = 0.30$ $[0.24, 0.37]$, LR 71.7, permutation $p = 0.001$, rewired $p = 0.015$. Gateway weighting added nothing beyond plain retaining relatedness (M3 vs M1: $p = 0.17$).

DTW trajectory classes ($k = 2$) were stable (bootstrap ARI 1.0), but HMM gave ARI = 0.094 against the DTW classes, so the two-class typology was not reproduced.

3.4 Evaluation 1: Stress-testing the gateway retention lead

Verdict: **FAILS.** The union-panel ΔAUC was $+0.001$ $[-0.012, +0.012]$. The D1 all- G -variant O1 gains ($+0.05$ – $+0.15$) were label-coverage artefacts (G : $+0.072 \rightarrow +0.002$ after controlling for coverage). The field intercept R^2 was 0.03 ($p = 0.55$) on the union panel.

3.5 Dataset 2: When research concepts were officially recognised

External-recognition lookup table for all 65,026 OpenAlex legacy concepts. Sources: MeSH 2026 (20,872 concepts), English Wikipedia (6,540 exact first revisions), Wikidata P571/P575 (1,425), ACM CCS 1998/2012 (1,298 concepts with events), MSC 2000/2010/2020 (1,121), PACS/PhySH (2,635), curated lists (589), JEL (213).

3.6 Research 1: Positioning study

Located 22 citable ANS papers with relation lines. Entry AUC comparators from Guevara et al. (2016): individuals 0.896, organisations 0.715, countries 0.682.

3.7 Review and hypothesis update

Reviewer score: 3 (blocking). Gateway retention was honestly disconfirmed, but several conclusions still contradicted evidence (“disconfirmed by all preregistered criteria” contradicted the positive within-field LPM; ordering was listed as “confirmed” despite negative lead-lag coefficients).

Hypothesis moved to **deepen**: headline shifted from gateway centrality (closed) to the retained-frontier lead from Experiment 6 ($d_0 = +0.281$). The RQ1 held-out indicator screen was made mandatory. Confidence decreased.

4 Iteration 3: Decisive frontier test and RQ1 held-out screen

4.1 Strategy and rationale

Iteration 3 delivered three linked claims: (1) a decisive held-out retained-frontier test against the RCA density rival, (2) the RQ1 53-indicator held-out screen, and (3) a record-repair evaluation. Five artifacts were commissioned; four completed and one failed.

4.2 Experiment 7: Do concepts spread from fields that keep them?

The decisive test on an independent frame (Experiment 5 minus every Experiment 6 concept: 11,841 concepts, 6,978 entry events in 6,076 informative strata).

Table 5: Retained-frontier field-entry results (iteration 3, Experiment 7). Conditional logit, held-out pooled. Source: `step2_heldout.json`.

Quantity	Value	95% CI
d_0 (held-out pooled, R3 vs R2)	+0.322	[+0.291, +0.355]
LR (R3 vs R2)	325.8	$p < 10^{-70}$
d_0 (S-strict, all rivals)	+0.304	[+0.268, +0.336]
d_0 per group: PHYS	+0.148	—
d_0 per group: LIFEENV	+0.401	—
d_0 per group: SOC	+0.297	—
d_0 per group: MATHDEC	+0.065	—
DL pooled (4 groups)	+0.243	[+0.118, +0.368]
I^2	0.92	—
Volume-matched contrast (R–N)	−0.028	[−0.105, +0.046]
Dose: 2/3/≥4 years	0.098/0.075/0.304	—
Min-cp proximity d_0	−0.021	$p = 0.012$

Frozen verdict: FRONTIER = PARTIAL. The volume-matched contrast was null (persistence confounded with volume), and under Hidalgo’s minimum-conditional-probability proximity d_0 vanished and reversed. The held-out dose response was not monotone (0.098/0.075/0.304). Abandonment: d_{lost} in A1 = −0.007 [−0.036, +0.022] (inconclusive); with d_0 it turned positive (+0.064 [+0.030, +0.095]).

4.3 Experiment 8: Which early network signals travel across fields

The RQ1 held-out deliverable: 53 indicators in 6 families (A co-occurrence ego-network 27, E popularity 6, F disciplinary spread 3, FR retained frontier 7, G gateway 7, S co-author 3), scored against 8 outcomes. Seven of 10 frozen indicators were confirmed for O_{2r_m50} (Table 6).

Learned models: ElasticNet combining all indicators added Spearman +0.059 [+0.046, +0.073] over B5 on held-out groups; EBM added +0.052 [+0.037, +0.067].

4.4 Evaluation 2: Auditing the record

Claims ledger: 246 rows (224 MATCH, 15 MISLABELLED, 6 MISMATCH, 1 FILE_FLAG_OVERRIDDEN; 58 blocking). External recognition (O5) was validated: pooled ρ with O_{2r_m50} was 0.014 [−0.045, +0.073] — unrelated to publication outcomes. 67% of concepts were recognised at or before t_0 .

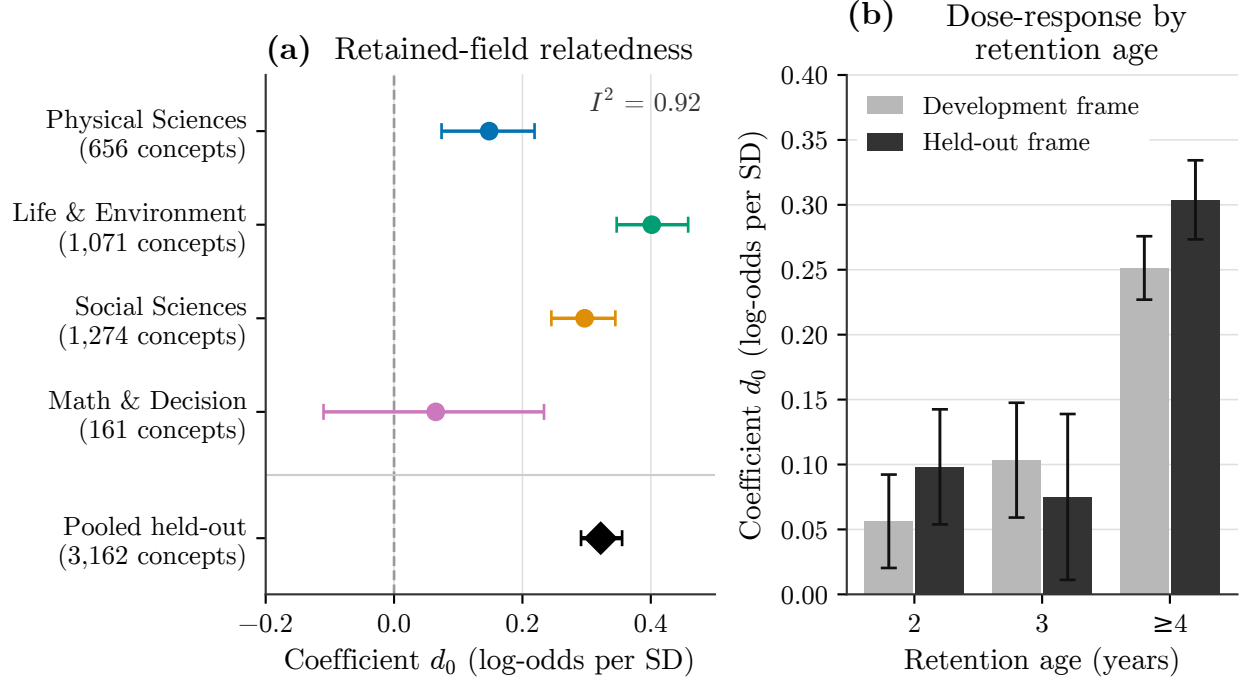


Figure 3: Retained-field relatedness predicts the next field a concept enters. (a) Conditional-logit coefficient d_0 (log-odds of entry per SD of relatedness to the off-home fields that currently retain the concept) on the held-out groups of the independent frame. Coloured circles are per-group estimates (Physical Sciences, 656 concepts; Life & Environment, 1,071; Social Sciences, 1,274; Math & Decision, 161), with 95% concept-bootstrap CIs. The black diamond is the pooled held-out estimate, $d_0 = 0.322$ [0.291, 0.355] (3,162 concepts, 6,978 entry events). The dashed line marks $d_0 = 0$. Three groups are clearly positive. The Math & Decision interval, the smallest group, crosses zero, and the groups differ substantially ($I^2 = 0.92$, DerSimonian–Laird over the four groups). (b) d_0 split by retention age (2, 3, ≥ 4 years), for the development frame (light grey, 4,302 concepts) and the held-out frame (dark grey, 3,162 concepts), with 95% Wald CIs (concept-clustered SE). In both frames the signal is largest for fields that have retained the concept for ≥ 4 years (0.251 and 0.304). The increase is monotone in the development frame only: on held-out data the 3-year estimate (0.075) falls below the 2-year one (0.098).

Table 6: Held-out indicator screen: the 10 indicators with highest DEV partial Spearman priority for rarefied breadth (O_{2r} , $m = 50$), tested on held-out groups. DerSimonian–Laird pooled estimates with 95% CI. Source: Experiment 8 (iteration 3), `heldout_summary.json`.

Indicator	Family	Pooled PSP	95% CI	I^2	Sign	Confirmed
M0_density_end	FR	+0.375	[+0.279, +0.462]	0.74	6/6	Yes
D_vol_end	FR	+0.307	[+0.256, +0.356]	0.10	6/6	Yes
CONTACT_REACH	FR	+0.211	[+0.161, +0.261]	0.00	6/6	Yes
n_{comm} (W3)	A	+0.167	[+0.063, +0.267]	0.78	6/6	Yes
NOV	A	+0.151	[+0.044, +0.255]	0.75	6/6	Yes
RETENTION_RATIO	FR	−0.114	[−0.160, −0.067]	0.00	6/6	Yes
ego_density (W3)	A	−0.102	[−0.151, −0.053]	0.00	6/6	Yes
Rao–Stirling ($\phi_{\min}$)	G	−0.072	[−0.153, +0.010]	0.44	5/6	No
G_{btw}	G	+0.056	[−0.006, +0.118]	0.33	6/6	No
log offhome vol.	F	−0.089	[−0.171, −0.007]	0.63	5/6	No

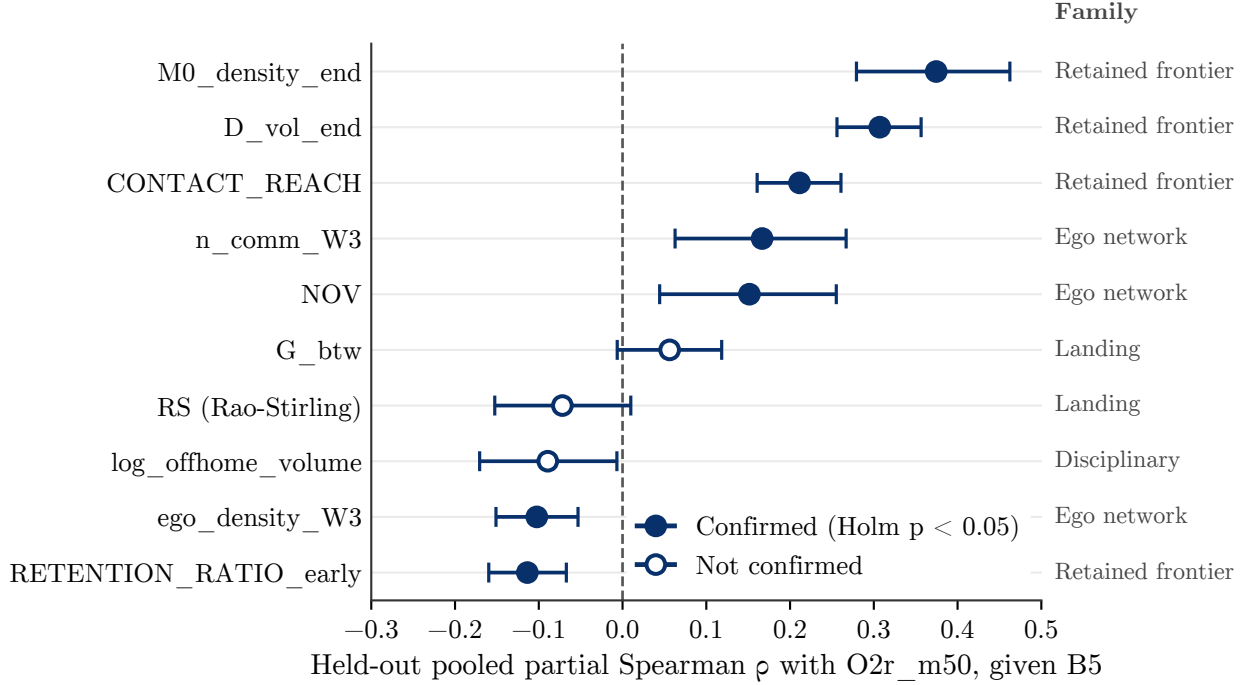


Figure 4: Held-out screen of the 10 indicators frozen on the DEV home groups (CS, Engineering, Biochemistry/Genetics, Medicine) for rarefied cross-field breadth (O_{2r_m50}). Each row shows the DerSimonian–Laird pooled partial Spearman ρ with O_{2r_m50} given the B5 baseline (x-axis). The pooling covers six units: four held-out home groups (Physical, Life/Environmental, Social, Math-/Decision sciences) and the 2010–14 onset cohort, split into DEV-home and other-home parts. Horizontal lines are 95% CIs. Filled circles mark indicators confirmed on held-out data (Holm $p < 0.05$ within the outcome family, pooled sign equal to the frozen DEV sign). Open circles mark indicators not confirmed. The dashed vertical line marks $\rho = 0$. Rows are ordered by pooled effect, and the right-hand column gives each indicator’s family. Seven of the ten are confirmed: five positive (M0_density_end, D_vol_end, CONTACT_REACH, n_comm_W3, NOV) and two negative (ego_density_W3, RETENTION_RATIO_early). M0_density_end and D_vol_end use cumulative field history up to $t_0 + 2$, so part of their signal predates onset.

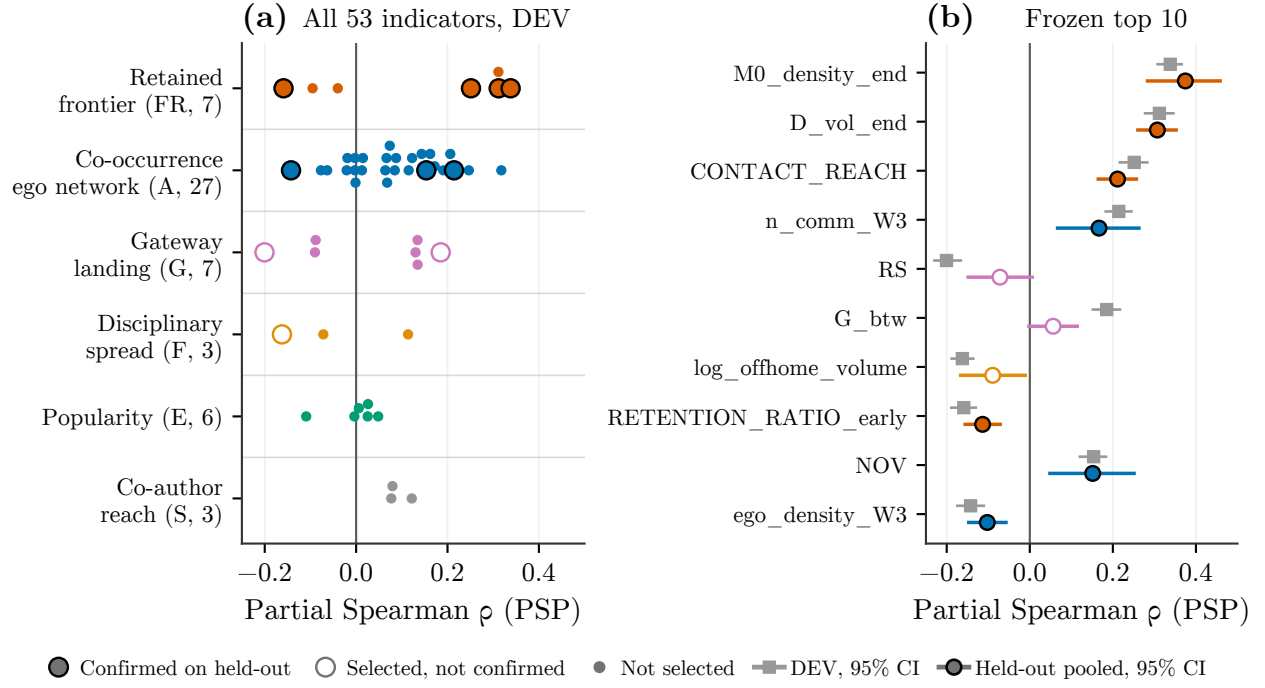


Figure 5: Complete indicator screen for rarefied cross-field breadth (O2r). (a) Partial Spearman ρ (PSP, controlling for rank(B5) plus group and onset-year dummies) of all 53 screened indicators on the DEV fields (CS, Engineering, BGM, Medicine), one row per family. Colour marks the family: retained frontier (FR), co-occurrence ego network (A), gateway landing (G), disciplinary spread (F), popularity (E) and co-author reach (S). Large filled circles are the 7 indicators confirmed on held-out fields; large open circles are the 3 selected on DEV but not confirmed; small dots are the 43 not selected. Selection kept indicators whose DEV 95% CI excluded zero with at most 30% missing, ranked them by $|\text{PSP}|$ and dropped near-duplicates ($|\rho| > 0.85$). This is why some high-ranking DEV indicators (e.g. D_rca_end, and D_rare, 88% missing) are small dots. (b) The frozen top 10 in DEV rank order. Grey squares are the DEV estimates with 95% concept-bootstrap CIs. Circles are held-out estimates pooled by DerSimonian–Laird over the PHYS, LIFEENV, SOC and MATHDEC groups, with 95% CIs. Filled circles are confirmed (Holm $p < 0.05$ with the frozen sign). All four retained-frontier (FR) and all three co-occurrence (A) selections replicate out of field. The gateway (G) and disciplinary-spread (F) selections shrink towards zero and are not confirmed. The held-out outcomes had been read once before by an earlier experiment. About half of the M0_density_end and D_vol_end effects reflects the concept’s pre-onset field footprint.

4.5 Research 2: Prior art for retained-field relatedness

Claim A (entry follows relatedness to retained fields) was judged **partially anticipated**. The closest science analogue was Cheng et al. (2023). No study found used a retained-only or duration-weighted density predictor.

4.6 Failed artifact

Experiment 9 (RQ2 trajectories on the 12,499-concept panel) failed with “output_format validation failed after 5 retries.” This was the entire RQ2 artifact for this iteration: contact×retention decomposition, trajectory typology, sequence tests and case studies.

4.7 Review and hypothesis update

Reviewer score: 3 (blocking). Key issues: Experiment 8 outcomes were mislabelled (O4 citation growth reported as transience), preregistered predictions misstated (new_edge_rate transfers: +0.118 [+0.072, +0.163], reversing iteration-1 dead end), the dose response was called monotone on DEV only, and none of Evaluation 2’s corrections were applied.

Hypothesis moved to **deepen**: headline shifted to openness vs consolidation, built from the Experiment 8 lead. The retained frontier was closed (volume-matched null, min-cp reversal). A fresh 2015–2016 onset cohort confirmation was mandated. Confidence decreased.

5 Iteration 4: Fresh-cohort openness test and RQ2 completion

5.1 Strategy and rationale

Iteration 4 attacked the Experiment 8 openness lead on four sides: replication on a never-screened 2015–2017 onset cohort; confound testing with concept type and pre-onset footprint; within-concept mechanism test; and boundary/specification analysis. Five artifacts were commissioned.

5.2 Experiment 10: Fresh-cohort OPEN confirmation

Single-unseal confirmation on 1,443 concepts (2015–2017 onsets, including 373 from the declared power extension to 2017). The OPEN index was the mean of six signed, z -scored ego-network components with constants frozen on the 12,499 Experiment 5 concepts.

Table 7: OPEN index control ladder on the 2015–2017 confirmatory cohort. PSP = partial Spearman priority with rarefied breadth (O_{2r} , $m = 50$), 95% concept-bootstrap CI. Source: Experiment 10 (iteration 4), `cohort_result.json`.

Build	R0	R1	R2	R3	R4	R5	n
OPEN _{home}	+0.12	+0.10	+0.09	+0.08	+0.07	+0.06	573
OPEN _{all}	+0.21	+0.18	+0.17	+0.17	+0.15	+0.14	630
OPEN _{size}	+0.18	+0.15	+0.15	+0.14	+0.12	+0.11	591

Frozen verdict: **CONFIRMED but marginal**. OPEN_{home} PSP = +0.091 [+0.013, +0.171] at R2, +0.080 [+0.001, +0.162] at R3. CIs included zero at R4/R5. DL pool over groups: +0.083 [−0.007, +0.173]. Practical prediction gain: +0.002 [−0.003, +0.008] — negligible.

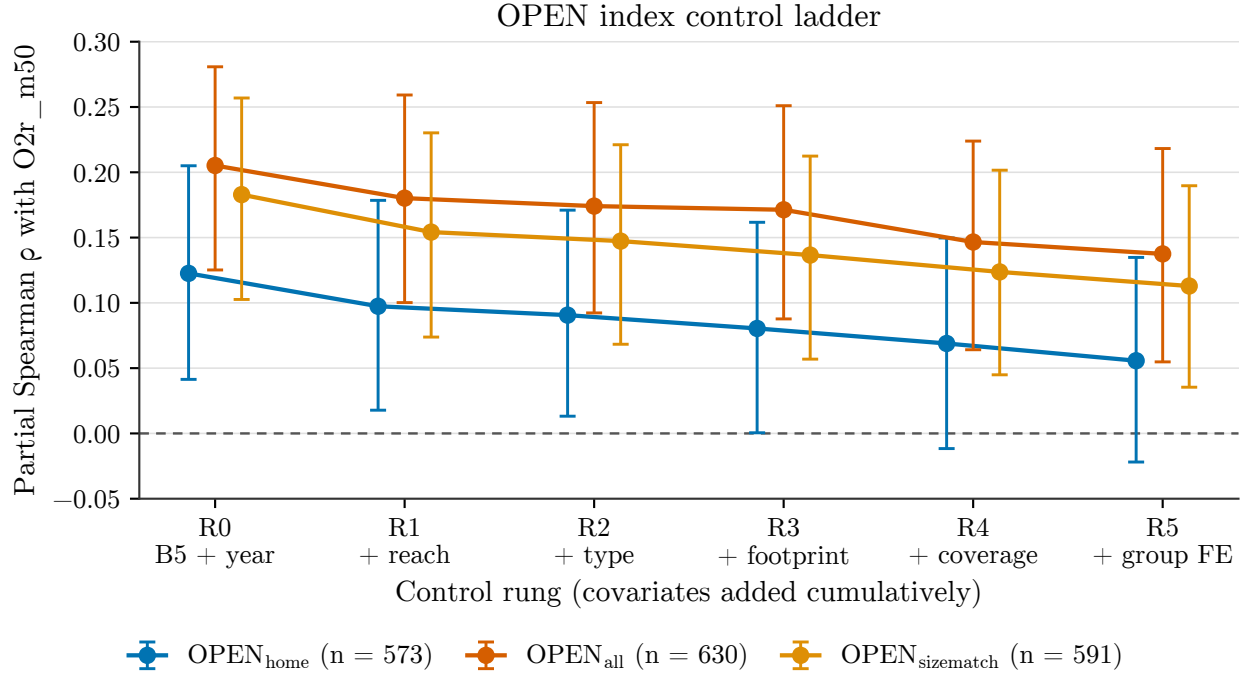


Figure 6: Control ladder for the early ego-network openness index (OPEN) on the fresh 2015–2017 onset cohort. The x-axis gives cumulative control rungs: R0 = B5 + onset year, R1 = + contact reach, R2 = + concept type, R3 = + pre-onset footprint, R4 = + label coverage, R5 = + group fixed effects. The y-axis gives the partial Spearman ρ between each OPEN build and rarefied off-home venue-field breadth ($O2r_{m50}$), given that rung’s covariates. Circles are point estimates; error bars are 95% concept-bootstrap CIs ($B = 2,000$); the dashed line marks $\rho = 0$. Colours mark the build: OPEN_{home} (blue, $n = 573$), OPEN_{all} (red, $n = 630$) and OPEN_{sizematch} (orange, $n = 591$); points are offset horizontally within each rung for legibility. All three estimates shrink as controls are added. The CIs of OPEN_{all} and OPEN_{sizematch} stay above zero at every rung, but OPEN_{all} is partly coupled to size mechanically. The home-only build is marginal: its CI touches zero at R3 and includes zero at R4 and R5.

The paired difference OPEN_{all} minus $\text{OPEN}_{\text{home}}$ at the footprint rung was $+0.093$ [$+0.016$, $+0.169$], confirming mechanical coupling: about half of the Experiment 8 signal was inflated by off-home papers entering the ego network.

5.3 Experiment 11 (incomplete): Within-concept closure test

Sealed a within-concept pre-registration and ran the DEV body models on 35,328 concept-years. Both preregistered effects were null: density PPML $b = -0.070$ [-0.180 , $+0.040$]; $\text{OPEN}_{\text{home}}$ $b = +0.015$ [-0.038 , $+0.069$]. By the frozen rule: **NOT SUPPORTED**. The Sun–Abraham event study was interrupted (OpenBLAS error); held-out and cohort were not run.

5.4 Experiment 12: Breadth decomposition and trajectories

RQ2 on all 12,499 concepts. The log-additive decomposition of the top-vs-bottom $O_{2r_{\text{resid}}}$ tercile gap:

Table 8: Breadth decomposition ($\log B_n = \log E_2 + \log M + \log \rho$): share of the top-versus-bottom tercile gap attributable to each component. Source: Experiment 12 (iteration 4), `decomposition_dev.json`, variant `i_pooled`.

Component	Share	95% CI
Exploration ($s_{E2} + s_M$)	0.732	[0.703, 0.764]
Early contact diversity (s_{E2})	0.779	[0.738, 0.818]
Retention (s_ρ)	0.268	[0.236, 0.297]
Difference (explore – retain)	0.464	[0.407, 0.528]

Trajectory typology: no discrete typology passed the naming rule (DTW $k = 4$ vs HMM $S = 5$: ARI 0.222; held-out re-cluster ARI 0.44/0.38). Verdict: **CONTINUUM**, with PC1 (38.8%) as a breadth-of-spread axis and PC2 (10.7%) as a keep-vs-lose axis.

5.5 Evaluation 3: Record fixes and openness robustness

Pre-onset footprint: about half of `M0_density_end` and `D_vol_end` was pre-onset. Post-onset `D_vol` dropped from $+0.317$ to $+0.176$ [$+0.114$, $+0.227$]; `M0_density_end` from $+0.374$ to $+0.187$ [$+0.145$, $+0.246$].

Specification curve across 1,920 specifications: 99.7% of pooled CIs above zero, permutation $p = 0.005$.

5.6 Research 3: Is “keep exploring, spread widest” new?

Openness $\rightarrow$ breadth was judged **partially anticipated** (Maillart 2026, Wang 2017, Weng 2013, Ugander 2012). Consolidation $\rightarrow$ less breadth was **partially anticipated in mechanism** and contradicted on other outcomes. Low retention ratio was judged **new**. Within-concept closure was judged **new as a lead-lag test**.

5.7 Review and hypothesis update

Reviewer score: 2 (blocking). Key issues: fabricated case-study rows in the report (five pairs did not exist in Experiment 12 output), Experiment 11 was missing from the record, several OPEN conclusions contradicted the artifact’s own reading of coupling.

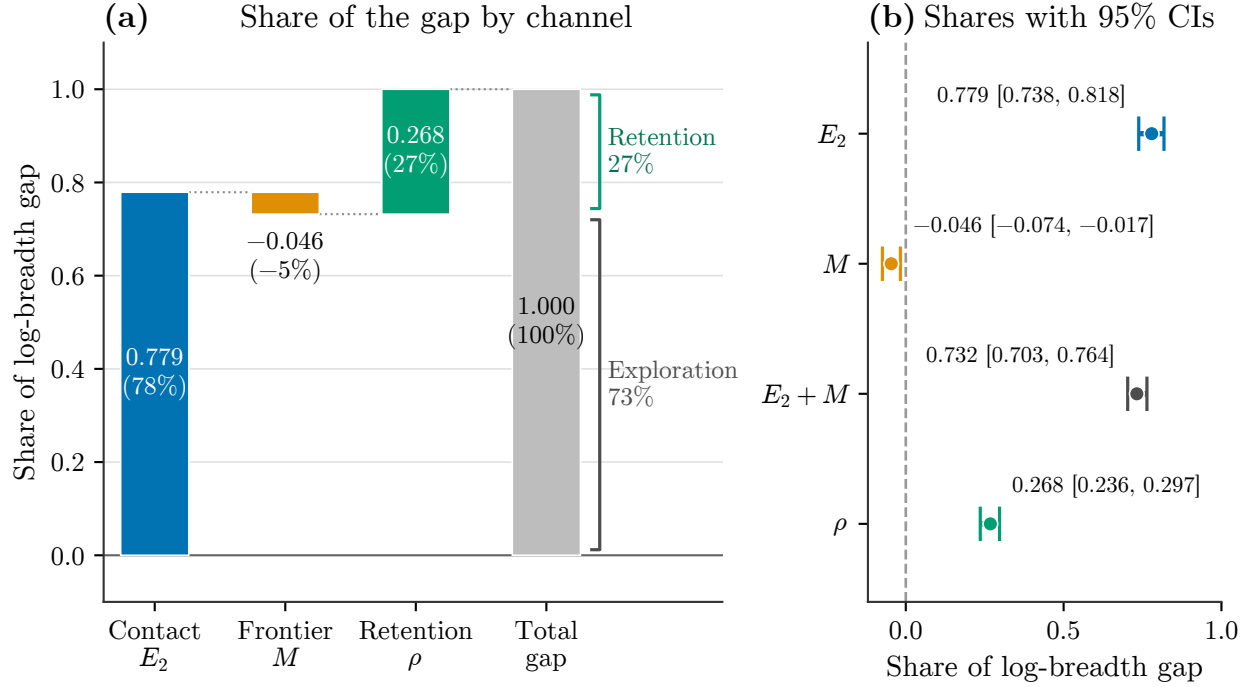


Figure 7: Log-additive decomposition of the gap in retained off-home breadth between the top and bottom terciles of size-adjusted breadth ($O2r_{\text{resid}}$). DEV concepts, pooled without volume strata, $n = 3,188, 1,063$ per tercile. (a) Waterfall of each channel's share of the log-breadth gap: early contact diversity E_2 (blue) contributes 0.779 (78%), frontier advance M (amber) is slightly negative at -0.046 (-5%), and retention ρ (green) contributes 0.268 (27%) up to the total gap (grey, 1.000). The bracket groups $E_2 + M$ as exploration (73%) and ρ as retention (27%). (b) The same shares, and the net exploration share $E_2 + M$ (dark grey), with 95% concept-bootstrap confidence intervals (2,000 resamples); the dashed line marks zero. Broad concepts start with wider contact; they do not mainly retain more fields later.

Hypothesis moved to **deepen**: sharpened from six-component openness to the decoupled home-only signal ($\text{NOV}_{\text{res}} +0.134$, edge persistence -0.112). Fresh confirmation on vocabulary-free Frame N concepts mandated. Confidence decreased.

6 Iteration 5: Vocabulary-free confirmation and mechanism

6.1 Strategy and rationale

The final iteration tested the home-only churn/novelty signal on a second, vocabulary-free population (Frame N) that no earlier step had scored; the frozen verdict was PARTIAL. It also tested the Cheng consistency reversal, decomposed the sampling-noise confound, and completed the Experiment 11 closure test. Five artifacts were commissioned, all completed.

6.2 Experiment 13: Does the churn signal hold for brand-new phrases?

Frame N: 636 vocabulary-free newborn title phrases (2003–2015 onsets), absent from the 56,643 legacy concepts. Pre-unseal power was 0.47. These newborns had 14% lower breadth and 89% higher transience than legacy concepts.

Frozen verdict: PARTIAL. $\text{OPEN}_{\text{home}}$ PSP = $+0.117$ [$+0.020$, $+0.218$] at R3 and $+0.086$ [-0.009 , $+0.190$] at R5. $\text{NOVCHURN}_{\text{home}}$ at R3: $+0.108$ [$+0.007$, $+0.211$]. NOV_{res} carried the signal ($+0.208$ [$+0.113$, $+0.303$]); edge persistence was null (-0.013). Coupling: ALL–HOME $+0.056$, not significant. Cheng consistency reversal was not confirmed on Frame N (-0.064 , CI included zero).

6.3 Experiment 14: Cheng’s consistency is a size effect

Rebuilt Cheng et al.’s (2023) ideational consistency on 12,311 concepts (105,839 concept-years). The negative-binomial twin reproduced their estimate: $b = 0.428$ ($+53.5\%$ /SD). Adding current volume $\log V(t)$ reduced it to $+1.3\%$ [$+0.5\%$, $+2.1\%$]; the A2/A1 ratio was 0.021 [0.009, 0.035]: about 98% of the effect was current size.

As an early trait, consistency predicted *narrower* cross-field reach: net of B5, the partial Spearman with $O_{2r.m50}$ was -0.069 [-0.093 , -0.047] (DL over 5 groups: -0.079 , $I^2 = 0$, 5/5 negative). Replicated on the 2015–2017 cohort: -0.111 [-0.197 , -0.030].

Within concepts, a more consistent year was followed by slightly *more* off-home entries ($b = +0.025$, CI [$+0.004$, $+0.048$]): the reach penalty is a between-concept trait.

6.4 Experiment 15: Why churning concepts spread; Experiment 11 completion

Part C (Experiment 11 completion): DEV verdict unchanged: NOT SUPPORTED. Held-out PPML: density $+0.068$ [-0.072 , $+0.209$]; $\text{OPEN}_{\text{home}}$ -0.079 [-0.146 , -0.013] — the opposite of the predicted sign. Cohort: both null. Sun–Abraham event study (DEV never-treated): lag 0–2 = -0.018 [-0.042 , $+0.004$], pre-trend $p = 0.52$. The within-concept closure test does not support a mechanism link from ego-network closure to entry slowdown.

Part A (partner decomposition): The home-only NOVCHURN signal comes from new-community partners ($C_2 = +0.102$ [$+0.069$, $+0.133$], Holm $p = 0.003$) that arrive through mixed-field papers ($C_4 = +0.103$ [$+0.071$, $+0.134$], Holm $p = 0.003$). Domain partners carry the signal (Shapley $\phi = +0.132$), not method partners ($\phi = +0.012$). Controlling for bridging-paper share halved the NOVCHURN partial from 0.118 to 0.056.

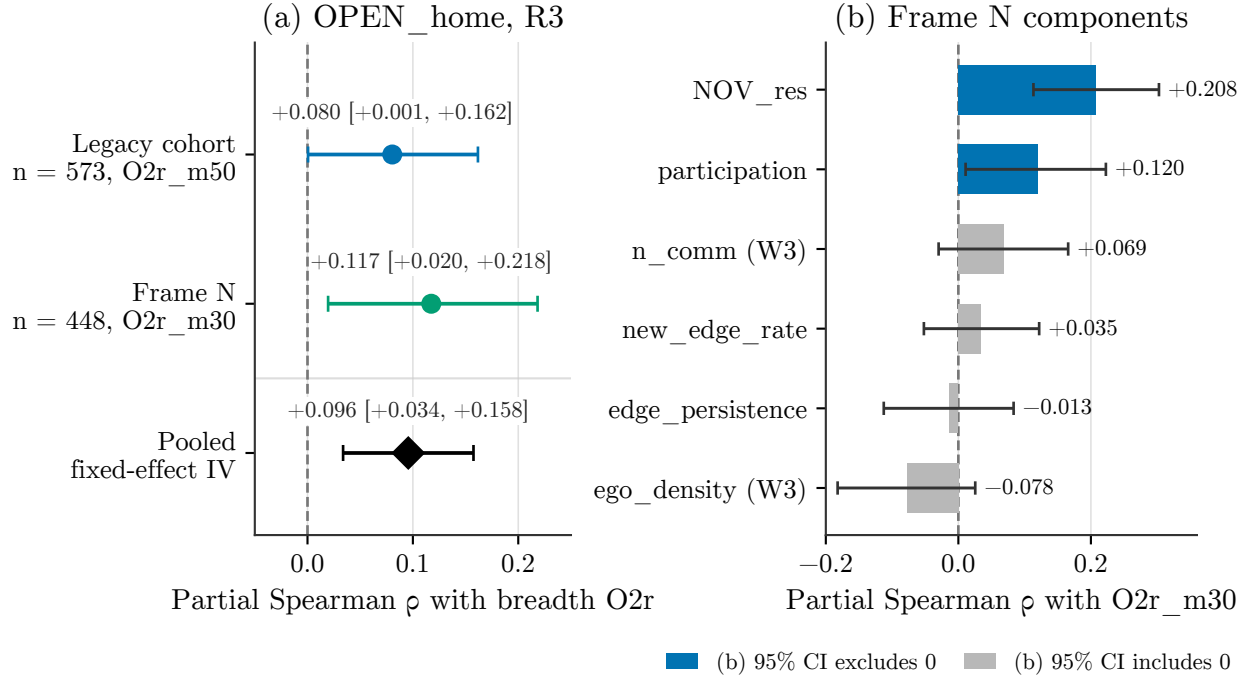


Figure 8: Confirmation of the OPEN signal on vocabulary-free Frame N concepts. (a) Forest plot of the OPEN_{home} partial Spearman ρ (PSP) with rarefied cross-field breadth at the footprint rung (R3), with 95% concept-bootstrap CIs: the 2015–17 legacy cohort (blue circle; $n = 573$, O2r_{m50}) gives +0.080 [+0.001, +0.162], Frame N (green circle; $n = 448$, O2r_{m30}) gives +0.117 [+0.020, +0.218], and their exploratory fixed-effect inverse-variance pool (black diamond) gives +0.096 [+0.034, +0.158]. The two bodies use different breadth rarefactions and are not identical rungs. (b) PSP with O2r_{m30} at R3 for the six OPEN_{home} components on Frame N ($n = 396$ –465 per component), with 95% CIs. Blue bars mark CIs that exclude 0 and grey bars CIs that include 0. NOV_{res} carries the signal (+0.208 [+0.113, +0.303]), participation is smaller (+0.120), and edge persistence is null (−0.013). Dashed lines mark zero.

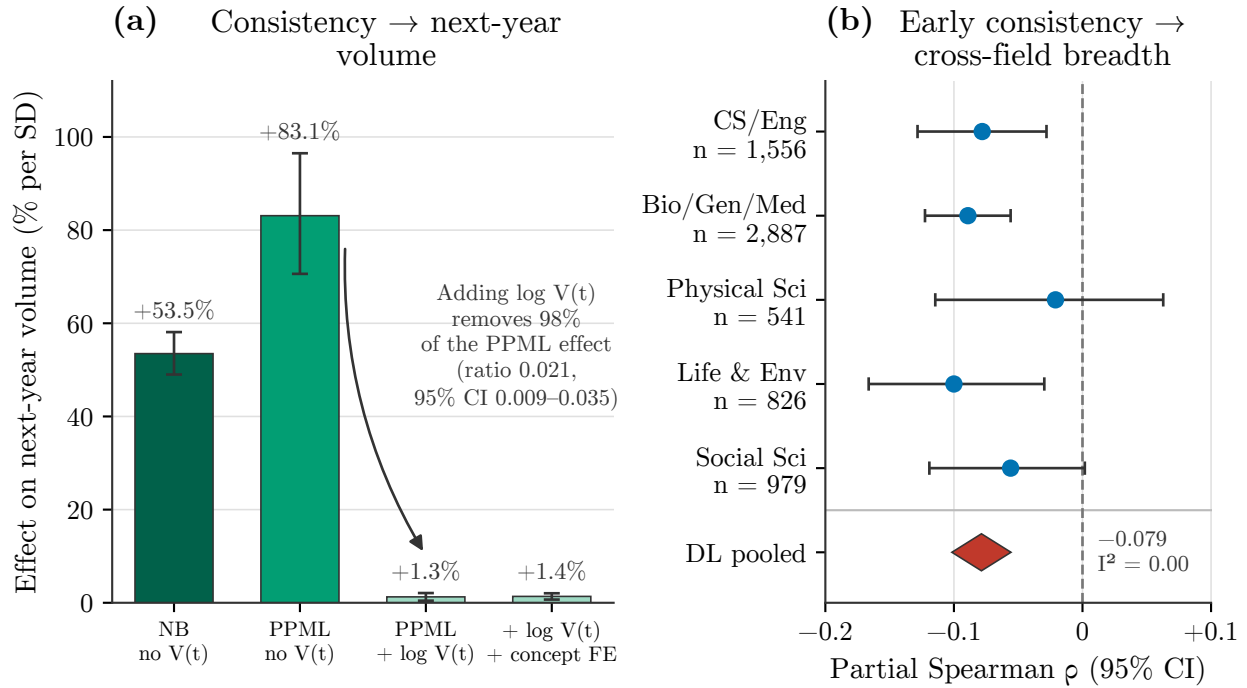


Figure 9: The consistency–breadth reversal (selection data). (a) Effect of ideational consistency (Cheng et al., 2023) on next-year volume, in % per SD, for 105,839 concept-years of 12,311 concepts. Dark green: the negative-binomial twin of Cheng’s specification, +53.5% ($b = 0.428$; model-based 95% CI). Mid green: PPML, +83.1%. Light green: PPML with current volume $\log V(t)$ added, +1.3% [+0.5, +2.1], and with concept fixed effects also added, +1.4%. PPML error bars are 95% CIs clustered by concept. The arrow marks the PPML ratio $A2/A1 = 0.021$ (500-draw concept-cluster bootstrap 95% CI 0.009–0.035): adding $\log V(t)$ removes about 98% of the effect. (b) Partial Spearman ρ between early consistency and rarefied cross-field breadth (O2r, $m = 50$), given the B5 baseline and onset-year, group and body dummies. Blue circles: five field groups with 95% concept-bootstrap CIs (n under each label). Red diamond: DerSimonian–Laird pooled estimate, -0.079 [$-0.102, -0.056$], $I^2 = 0$. The dashed line marks $\rho = 0$. All five point estimates are negative; the Physical Sci and Social Sci intervals include 0.

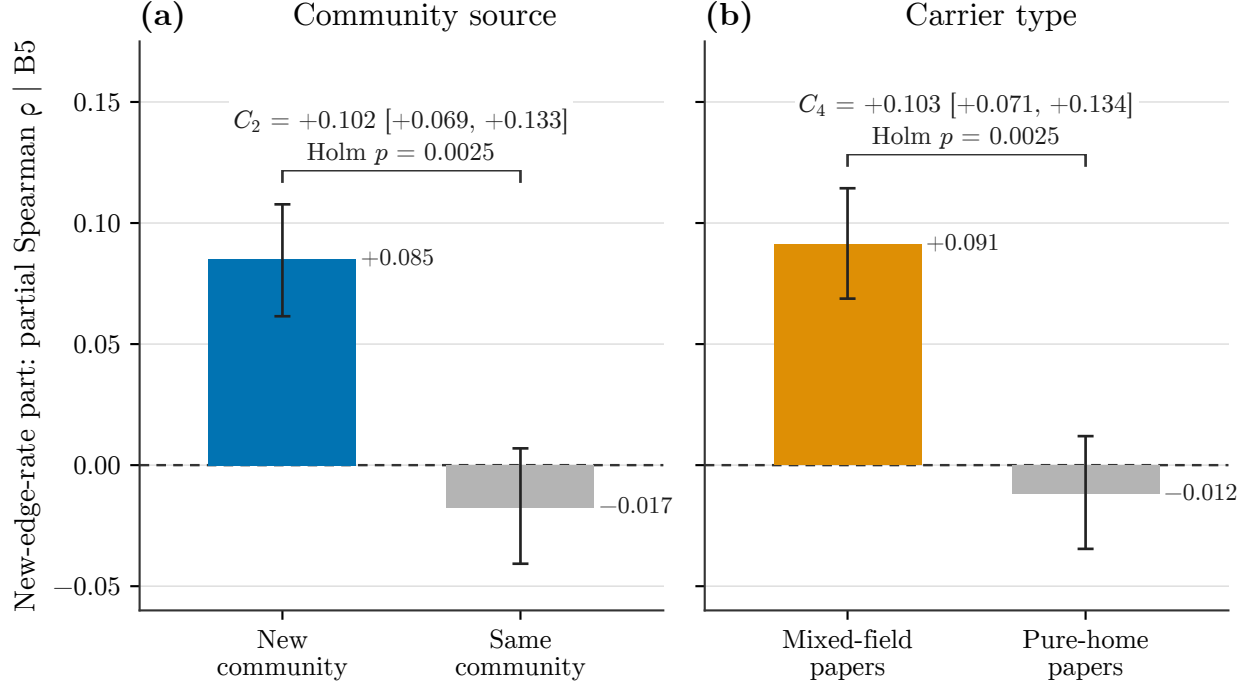


Figure 10: Which co-occurrence partners carry the openness–breadth signal? Each bar is one partner class’s part of a concept’s early home new-edge rate, scored as its partial Spearman ρ with cross-field breadth ($O2r_{m50}$) given B5 plus onset-year, group and body dummies, pooled over the EXP5 frame (DEV, old held-out and 2010–14 cohort; $n = 7,203$ concepts). Whiskers are 95% concept-bootstrap intervals (2,000 draws); the dashed line marks $\rho = 0$; each bracket gives the contrast between the two bars with its 95% interval and Holm-adjusted p over five pre-declared contrasts. (a) Partners from new Leiden communities (blue, +0.085) carry the signal; same-community partners (grey, -0.017) do not ($C_2 = +0.102$ [$+0.069, +0.133$], Holm $p = 0.0025$). (b) Partners arriving through mixed-field papers (orange, +0.091) carry it; partners from pure-home-field papers (grey, -0.012) do not ($C_4 = +0.103$ [$+0.071, +0.134$], Holm $p = 0.0025$). The small unclassified-community part (+0.011) is not drawn. Exploratory decomposition on selection data; pooled over the four held-out groups, C_4 is +0.060 [$-0.003, +0.123$].

Part B (trait stability): $\text{OPEN}_{\text{home}}$ split-half reliability was 0.49, NOVCHURN 0.26–0.29 (rising to 0.51–0.57 on window retest and about 0.50 at degree ≥ 5). Disattenuated retest 0.86–0.91.

6.5 Experiment 16: Is neighbourhood churn real or thin-sample noise?

On 13,444 selection concepts. The within-concept year-label permutation excess (V2) cut NOVCHURN from +0.116 [+0.09, +0.14] to +0.008 [−0.02, +0.03], retaining only 6.5%. The R^2 of raw persistence on the V2 null mean was 0.66 — two-thirds of between-concept persistence variance is explained by the static topic mix of the concept’s home neighbourhood.

Verdict: PARTLY_THIN. The signal is a static topical-dispersion property, not temporal partner turnover. Fixed- n rarefaction at 10 papers/year retained 68% of the pooled association. The degree-preserving configuration-null z -score retained 100% and raised $\text{OPEN}_{\text{home}}$ from +0.092 to +0.115. But the V2 excess variants had split-half reliability of only 0.01–0.05, so V2 could not adjudicate temporal churn at typical sample sizes.

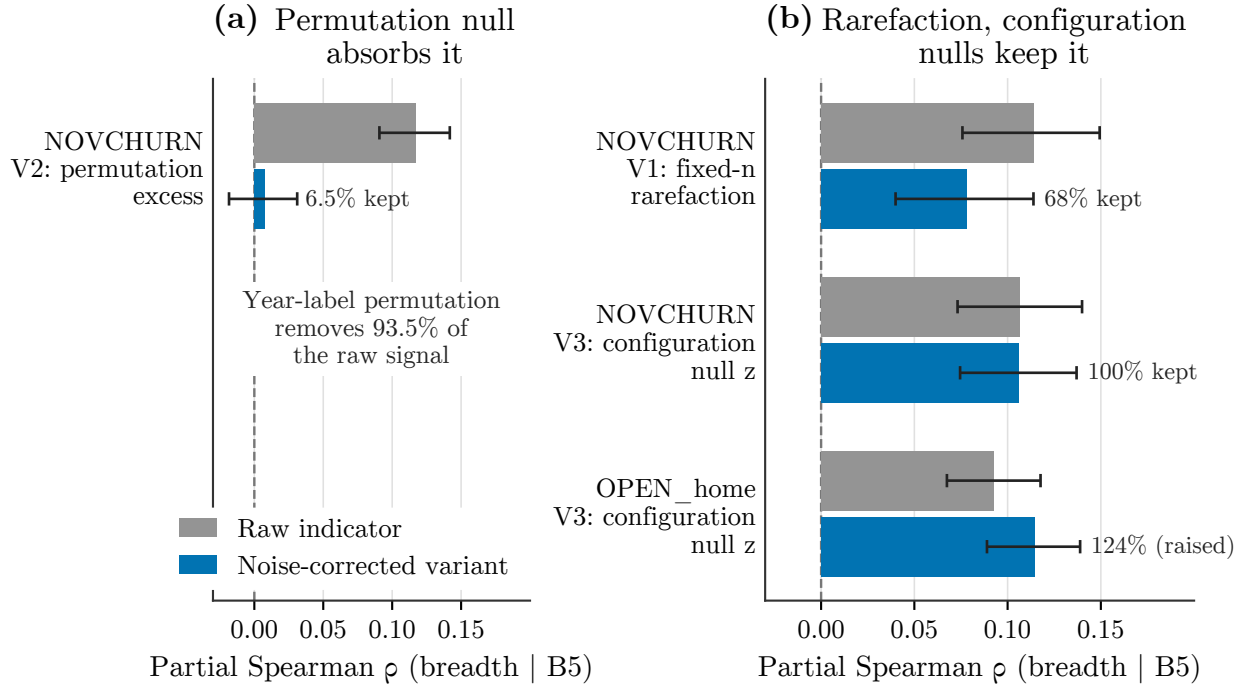


Figure 11: The churn/openness signal behaves like topical non-redundancy of the home neighbourhood, not year-to-year partner turnover. Bars show the pooled partial Spearman ρ between each indicator and later cross-field breadth (O2r_{m50}), given B5 and body dummies (rung R2). Grey bars are the raw indicator and blue bars the noise-corrected variant, computed on the same concepts. Whiskers are 95% percentile CIs from 2,000 concept-level bootstrap resamples, the dashed line marks $\rho = 0$, and the labels give the corrected-to-raw ratio. (a) The within-concept year-label permutation excess (V2) cuts NOVCHURN from +0.117 [+0.091, +0.142] to +0.008 [−0.018, +0.031] ($n = 6,203$), keeping 6.5%. (b) Fixed- n rarefaction at 10 papers per year (V1) keeps 68% of NOVCHURN (+0.114 $\rightarrow$ +0.078, $n = 2,872$). The configuration-null z -score (V3) keeps 100% of NOVCHURN (+0.106 $\rightarrow$ +0.106, $n = 3,993$) and raises $\text{OPEN}_{\text{home}}$ to 124% (+0.092 $\rightarrow$ +0.115, $n = 6,322$). These are selection data (outcomes previously unsealed): robustness evidence, not confirmation.

Disattenuated pooled NOVCHURN_{raw} PSP: 0.178 [0.141, 0.214].

6.6 Evaluation 4: Record repair and openness evidence pool

Claims ledger v4: 1,769 rows, 0 MISMATCH, 0 NOT_FOUND, 0 orphans. All ten MUST-FIX items cleared.

Evidence synthesis (descriptive, R2 rung, O_{2r-m50} , DL on Fisher z with HKSJ): OPEN_{home} non-selection pool ($k = 6$: 4 held-out groups + 2010–14 cohort + 2015–17 cohort): +0.069 DL [+0.038, +0.100], HKSJ [+0.042, +0.096], $I^2 = 0$, 6/6 positive. DEV selection body +0.109, shrinkage ratio 1.58. NOVCHURN_{home} ($k = 5$): +0.105 [+0.069, +0.140].

6.7 Review

Reviewer score: 2 (blocking). The document submitted was the paper manuscript, not the chronological notebook. The chronological record with iterations 1–4 existed on disk (Evaluation 4’s corrected copy with a verified ledger) but was not carried forward, and no iteration-5 section was written. The headline numbers from executed artifacts were correct, but several conclusions contradicted the run’s own artifacts (the abstract’s +0.17 was OPEN_{all}, which Experiment 10 called mechanically coupled; Frame N was called a “confirmation” but its frozen verdict was PARTIAL).

7 What the run learned

7.1 Confirmed findings

1. **Seven early indicators predict cross-field breadth beyond B5** (Experiment 8, iteration 3). The confirmed indicators span two families: retained frontier (M0_density_end, D_vol_end, CONTACT_REACH, RETENTION_RATIO) and co-occurrence ego-network topology (n_{comm} , NOV_{res}, ego_density). About half of the top two is pre-onset footprint.
2. **The OPEN index captures early neighbourhood openness** (Experiment 10, iteration 4; Evaluation 4, iteration 5). The Eval4 descriptive pool of OPEN_{home} over 6 non-selection bodies gives partial Spearman +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive. The home-only signal is marginal (DL CI includes zero on the fresh cohort), with no practical forecasting gain (+0.002).
3. **Concepts spread to fields related to those retaining them** (Experiment 7, iteration 3). $d_0 = 0.322$ [0.291, 0.355] on the independent frame, extending the principle of relatedness. However, the volume-matched contrast is null, and the effect reverses under min-cp proximity.
4. **Consistency predicts volume but narrower breadth** (Experiment 14, iteration 5). Cheng et al.’s consistency +53.5%/SD for volume drops to +1.3% once size is controlled. As an early trait, consistency predicts *less* breadth: -0.069 [-0.093 , -0.047], $I^2 = 0$, 5/5 negative.
5. **Breadth is 73% early exploration, 27% retention** (Experiment 12, iteration 4). Broad concepts start with wider contact, not by advancing a wider frontier ($s_M = -0.046$).
6. **The signal is topical non-redundancy, not temporal turnover** (Experiment 16, iteration 5). A within-concept permutation null absorbs the year-to-year component while the signal persists.

7.2 Dead ends

1. **Citation naturalisation gap** A_h^* : null ($\Delta\rho = -0.006$, iteration 1). Medians were negative in all groups.
2. **Gateway field retention**: disconfirmed ($\Delta\text{AUC} = -0.00001$, iteration 2). Gateway was a domain-specific proxy for “fields that keep things.”
3. **Social reach (candidate S)**: tested in iteration 3, not confirmed for any outcome.
4. **Gateway weighting for field entry**: adds nothing beyond plain retaining relatedness (iteration 2).
5. **Discrete trajectory typology**: failed naming rule (DTW vs HMM ARI = 0.222, iteration 4). Breadth is a continuum.
6. **Within-concept closure $\rightarrow$ entry slowdown**: null on DEV ($b = -0.070$, $+0.015$; iteration 4), opposite sign on held-out (iteration 5).
7. **Ordering (gateway first $\rightarrow$ take-off)**: mixed. Negative lead-lag, pre-trend, reverse path significant on DEV.
8. **External recognition (O5)**: unrelated to publication outcomes ($\rho = 0.014$, iteration 3).
9. **Abandonment penalty**: inconclusive (specification-dependent; iterations 3–5).

7.3 What is still open

1. **Causal mechanism**: the run shows association, not causation. Early openness could reflect intrinsic generality, community diversity, or problem breadth.
2. **Domain heterogeneity**: Life & Environment shows the weakest OPEN signal (PSP $+0.071$ vs $+0.186$ for others pooled); the retained-field relatedness effect has $I^2 = 0.92$.
3. **Temporal turnover vs static dispersion**: V2 excess has split-half reliability below 0.05, so the question cannot be adjudicated at typical sample sizes. Temporal turnover is untested rather than refuted.
4. **Vocabulary survivorship**: Frame N was small (636 concepts) with pre-unseal power of 0.47.
5. **Measurement noise**: split-half reliability of $\text{OPEN}_{\text{home}}$ is 0.49. Individual-concept prediction is imprecise.

8 Artifact inventory

Table 9: All artifacts commissioned across five iterations.

Iter.	Name	Type	Status	Title
1	Experiment 1	experiment	completed	Does citing a concept ‘as your own’ predict its spread?
1	Experiment 2	experiment	failed	Do independent author groups predict spread?
1	Experiment 3	experiment	completed	Do diverse topic ties predict concept spread?
1	Experiment 4	experiment	completed	Where a concept lands early vs how broadly it spreads
1	Dataset 1	dataset	failed	Sealed test set of new science concepts
2	Experiment 5	experiment	completed	Do hub fields keep new concepts? Held-out test
2	Experiment 6	experiment	completed	Where new scientific concepts spread next
2	Evaluation 1	evaluation	completed	Does the gateway-field retention signal replicate?
2	Dataset 2	dataset	completed	When research concepts were officially recognised
2	Research 1	research	completed	How our results compare with related papers
3	Experiment 7	experiment	completed	Do concepts spread from fields that keep them?
3	Experiment 8	experiment	completed	Which early network signals of new topics travel
3	Experiment 9	experiment	failed	How new concepts spread: paths and reasons
3	Evaluation 2	evaluation	completed	Auditing the record before the paper
3	Research 2	research	completed	Is ‘fields that keep it’ new? Prior art and venue check
4	Experiment 10	experiment	completed	Do open-neighbourhood concepts spread? Fresh-cohort test
4	Experiment 11	experiment	incomplete	Does closing up at home slow a concept’s spread?
4	Experiment 12	experiment	completed	How concepts spread: early reach vs keeping fields
4	Evaluation 3	evaluation	completed	Record fixes and openness robustness tests
4	Research 3	research	completed	Is ‘keep exploring, spread widest’ already known?
5	Experiment 13	experiment	completed	Does the churn signal hold for brand-new phrases?
5	Experiment 14	experiment	completed	Cheng’s consistency: size effect, not reach
5	Experiment 15	experiment	completed	Why churning concepts spread; Exp11 test completed
5	Evaluation 4	evaluation	completed	Record repair and openness evidence pool
5	Experiment 16	experiment	completed	Is research-topic churn real or small-sample noise?